중앙대학교
CHUNG-ANG UNIVERSITY

VIDEOMIND :

A CHAIN-OF-LORA AGENT FOR TEMPORAL-GROUNDED VIDEO REASONING

2026-05-21

Task 설명

Audio Moment Retrieval from Long Audio



Audio moment retrieval is the task of retrieving specific moments within long audio recordings that align with a given textual query. While conventional language-based audio retrieval aimed to accurately capture the correspondence between queries and audio clips, this task focuses on localizing exactly which part of a long audio recording corresponds to the query. The primary challenge is to capture temporal contexts within long audio, which requires effective sequence modeling and learning methods to enhance retrieval accuracy.

Organizers



Hokuto Munakata
LY Corporation



Tatsuya Komatsu
LY Corporation



Keisuke Imoto
Kyoto University



Taichi Nishimura
Sony Interactive Entertainment



Paul Primus
Johannes Kepler University Linz



Huang Xie
Tampere University



Tuomas Virtanen
Tampere University

I. INTRODUCTION

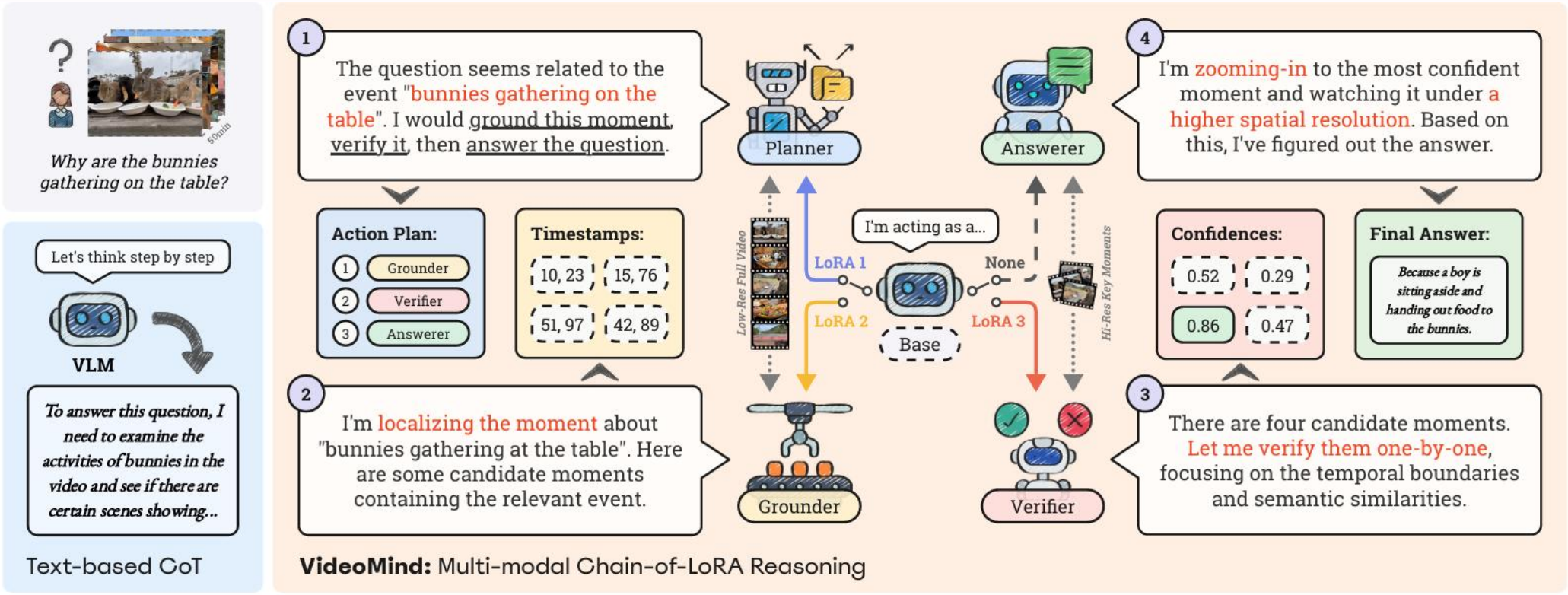
1. 기존 VLM 모델의 한계

- 정적 이미지 추론에는 강점을 보임
- 긴 비디오에서의 시간적 흐름 파악 불가
- 초기 시점의 정보 회수 및 재검증 실패

2. VideoMind

- 인간이 긴 비디오를 시청하고 추론하는 자연스러운 인지 과정을 4단계의 에이전트 워크플로우로 구현
- 역할 기반 에이전트 워크 플로우 도입
- 선형적 텍스트 처리에서 벗어나, 시작 중심의 반복적이고 모듈화된 인지 프로세스 구축

II. VideoMind



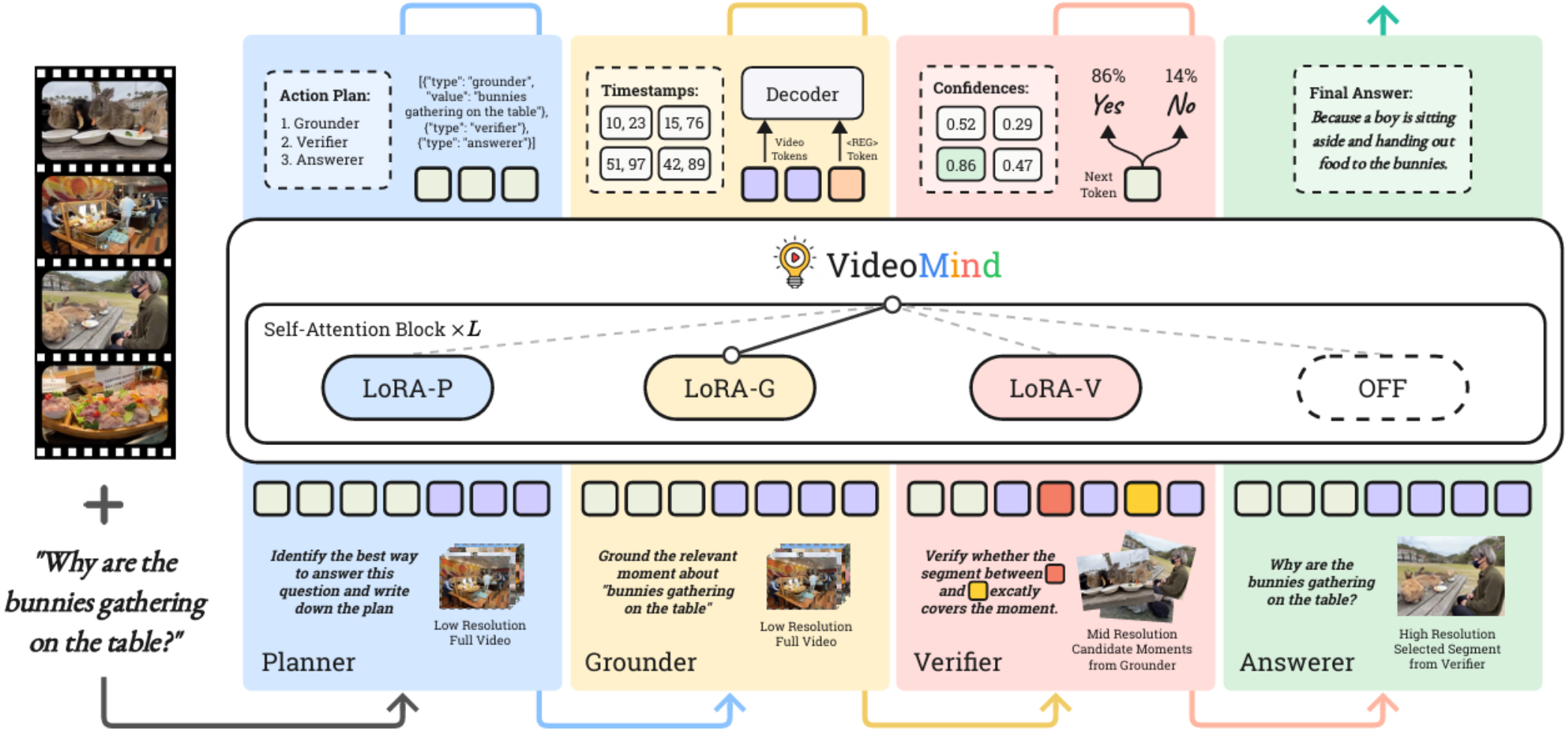
1. Planner : 질문을 분석하여 역할들의 호출 순서를 동적으로 조율

2. Grounder : 관련 비디오 moment을 timestamp 형태로 정확히 특정

3. Verifier : 후보 순간들을 zoom-in하여 신뢰도 평가 및 필터링

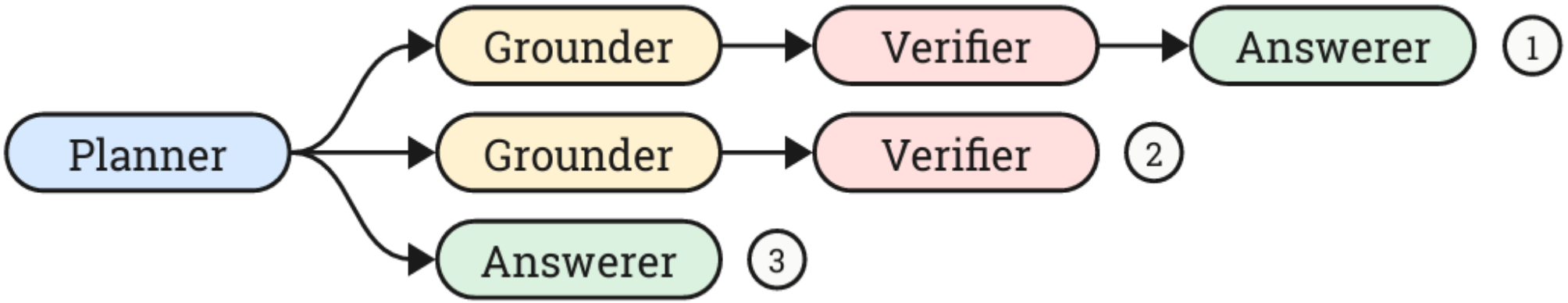
4. Answerer : 최종 확정된 비디오 조각을 바탕으로 자연어 답변 생성

II. VideoMind



* VideoMind는 video, query에 맞춰 Planner부터 Answerer까지 개별 모듈을 호출하여 단계적인 추론을 수행

I. Planner




```
[
    {"type": "grounder"},
    {"type": "verifier"},
    {"type": "answerer"}
]
```

1. Grounding Video QA : 정확한 시간대와 답변이 모두 필요한 경우

2. Video Temporal Grounding : 특정 사건의 시간대만 찾는 경우

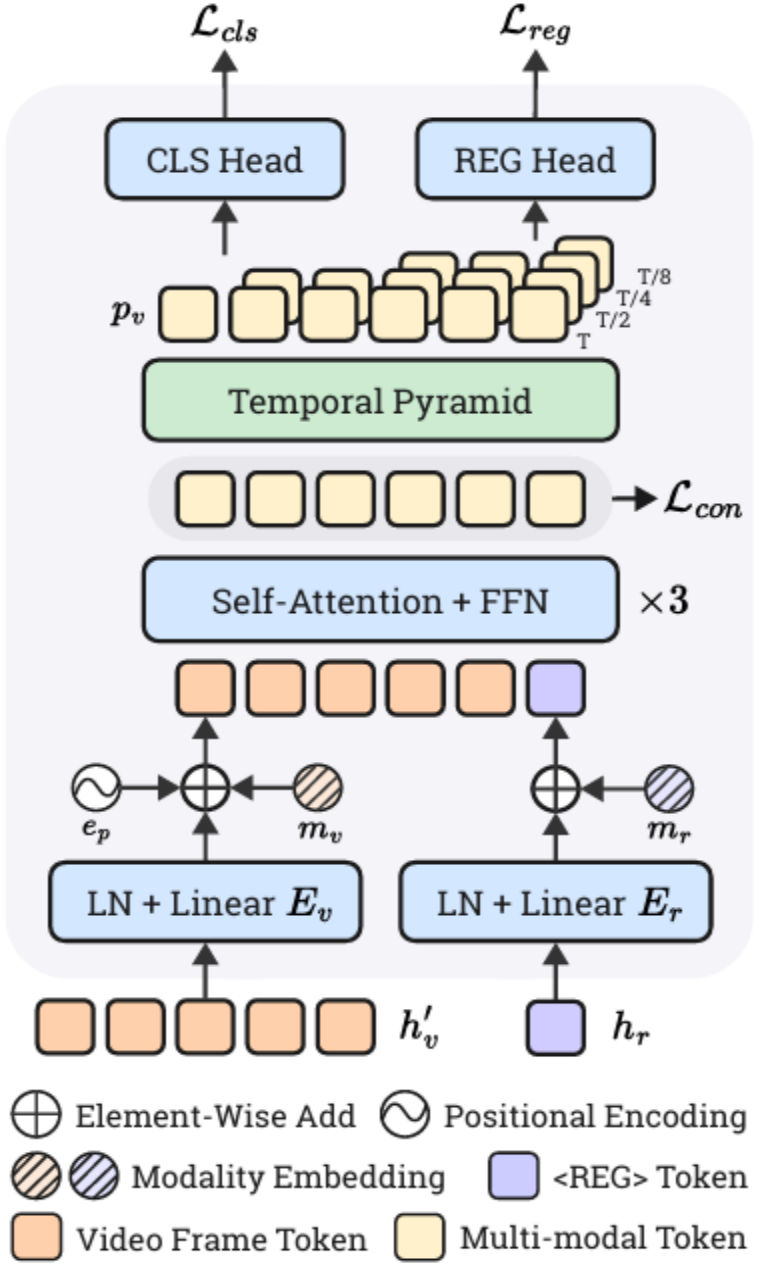
3. General VideoQA : 전체 비디오의 맥락적 이해만으로 충분한 경우

II. Grounder : Timestamp Decoder

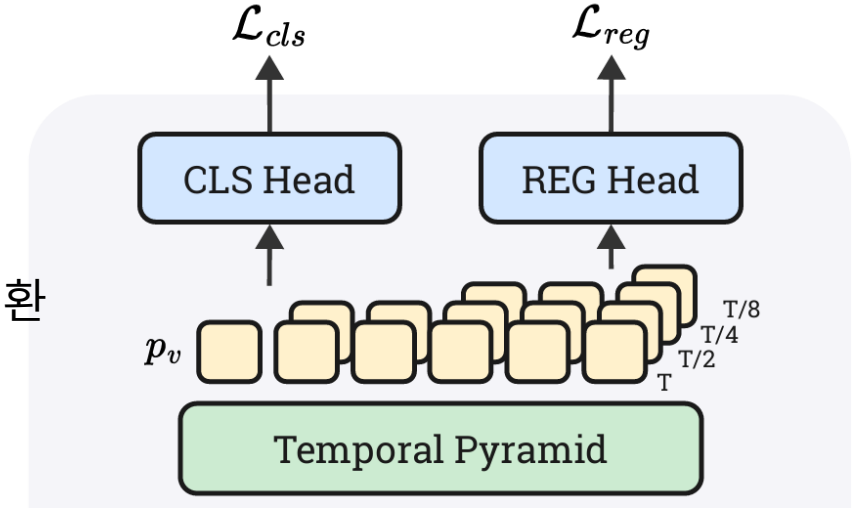
$$\mathbf{h}'_v = \text{AvgPool}(\mathbf{h}_v) \in \mathbb{R}^{T \times D_L}$$

$$\mathbf{e}_v = E_v(\mathbf{h}'_v) \in \mathbb{R}^{T \times D}, \quad \mathbf{e}_r = E_r(\mathbf{h}_r) \in \mathbb{R}^{1 \times D}$$

$$[\mathbf{e}'_v; \mathbf{e}'_r] = \text{Transformer}([\mathbf{e}_v + \mathbf{m}_v + \mathbf{e}_p; \mathbf{h}_r + \mathbf{m}_r])$$



II. Grounder : Temporal Feature Pyramid & Prediction Head



1. Temporal Feature Pyramid

- Contextualized frame embedding e'_v를 multi-scale temporal representation으로 변환
- Conv1D → LayerNorm → SiLU block 사용 4개 temporal scale 생성
- 4개의 temporal scale 생성 : T, T/2, T/4, T/8

2. Prediction Head

Head	역할	출력
CLS Head	각 frame이 target moment 내부인지 판단	confidence score $\hat{c}_i$
REG Head	start/end boundary offset 예측	$[b^{is}, b^{ie}]$

II. Grounder : loss_function

1. Classification loss

- Hyperparameter : α = 0.9 and γ = 2.0

$$\mathcal{L}_{cls} = -\lambda_{cls}\alpha(1-\hat{c}_i)^{\gamma}\log(\hat{c}_i)$$

2. Boundary regression loss

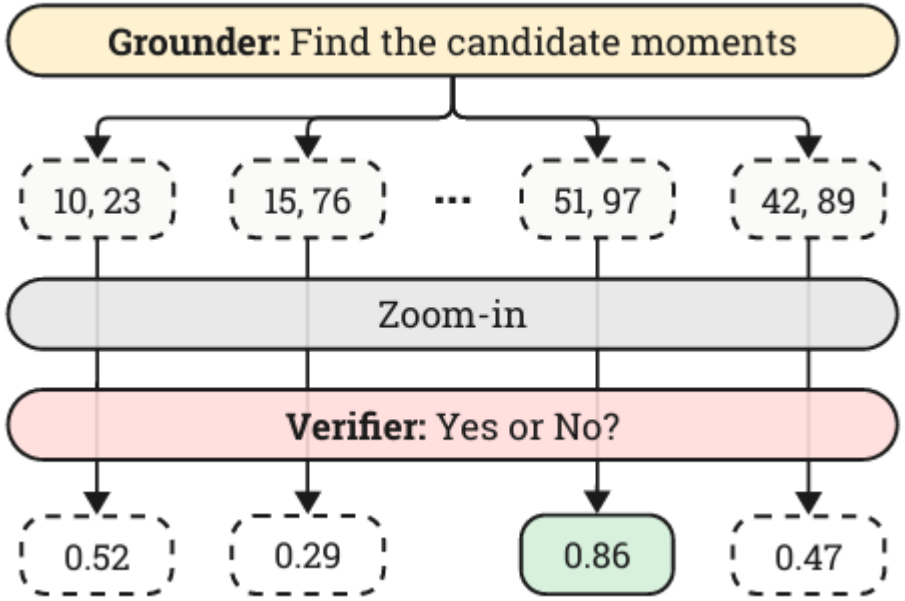
$$\mathcal{L}_{reg} = \lambda_{reg}(|b_i^s - \hat{b}_i^s| + |b_i^e - \hat{b}_i^e|)$$

3. Contrastive loss

$$\mathcal{L}_{con} = -\lambda_{con}\log\frac{\exp(s_p/\tau)}{\exp(s_p/\tau) + \sum_{i\in\Theta}\exp(s_i/\tau)}$$

4. λcls = 5.0, λreg = 1.0, λcon = 0.05.

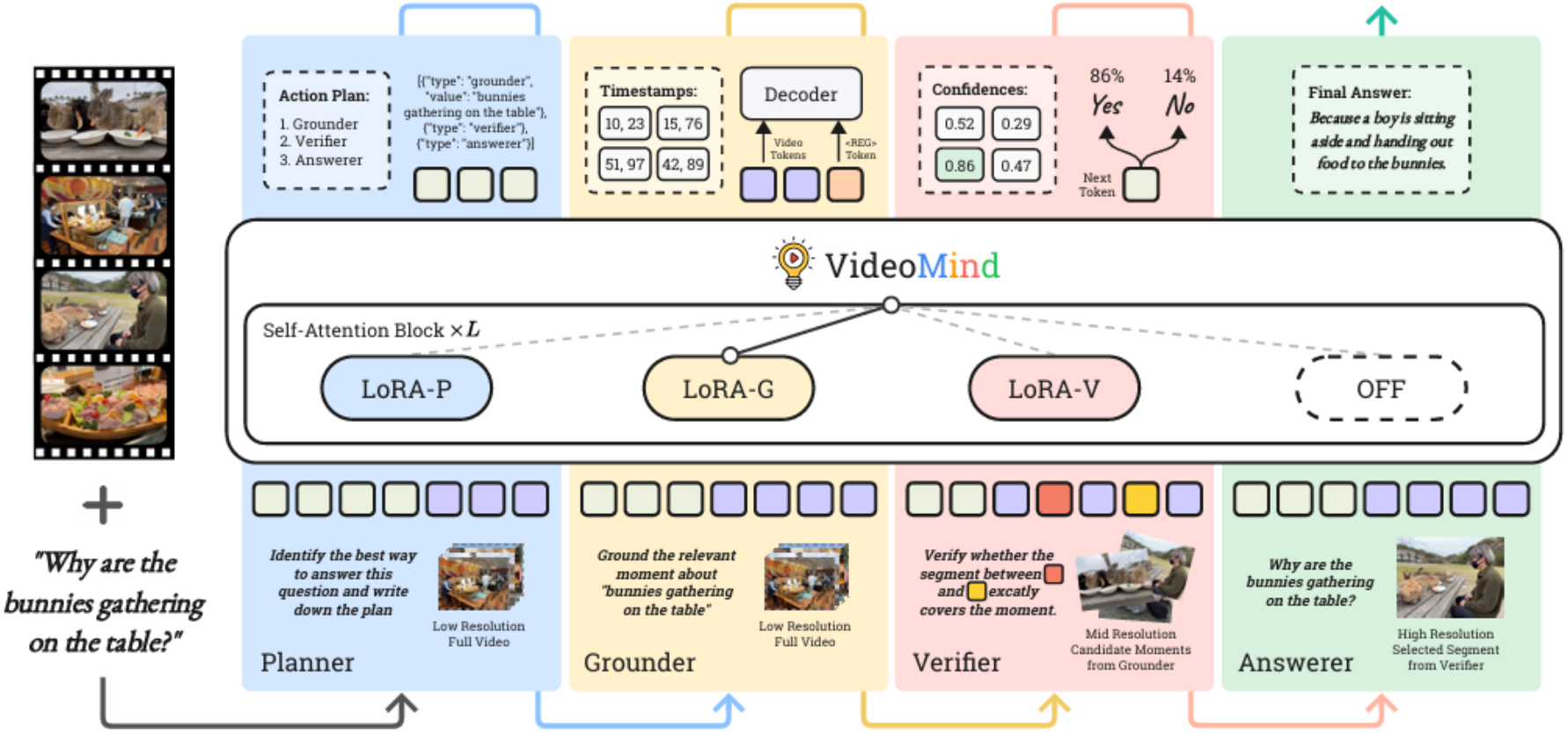
III. Verifier



- Grounder는 top-5 candidate moments 생성
- 각 candidate를 50% 확장 후 crop Verifier가 해당 구간이 query와 정확히 맞는지 Yes/No 판단
- <SEG-START>, <SEG-END> token으로 moment boundary 표시
- 최종 score: Sigmoid(Ly - Ln)

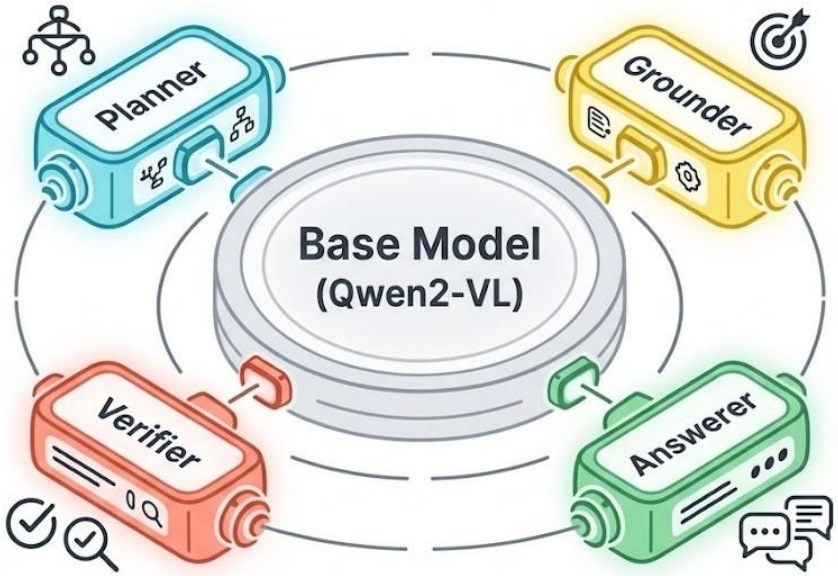
III. Method

IV. Answerer



- 사전 검증된 Cropped Segment에만 집중하여 hallucination 최소화
- LLM의 강력한 추론 능력을 훼손 없이 그대로 활용
- 시각적 증거와 완벽히 정렬된 텍스트 응답 생성

V. Chain of LoRA



단일 베이스 모델 + 4개의 LoRA 어댑터 (메모리 최적화)

- 별도 fine-tuning 없이 original LLM(Qwen2-VL) 사용
- Planner, Grounder, Verifier는 각각 role-specific LoRA 사용
- 하나의 base LLM에 여러 LoRA adapter를 연결
- inference 중 필요한 역할의 LoRA만 활성화

IV.Experiments

I. Grounded VideoQA Performance

Table 2: Performance comparison on Grounded VideoQA on CG-Bench (Chen et al., 2024a).

Method	Size	long-acc.	mIoU	rec.@IoU	acc.@IoU
GPT-4o (OpenAI, 2024a)	–	**45.2**	5.62	8.30	4.38
Gemini-1.5-Pro (Reid et al., 2024)	–	37.2	3.95	5.81	2.53
Claude-3.5-Sonnet (Anthropic, 2025)	–	40.5	3.99	5.67	2.79
Video-LLaVA (Lin et al., 2023a)	7B	16.2	1.13	1.96	0.59
VideoLLaMA (Zhang et al., 2023b)	7B	18.4	1.21	1.87	0.84
VideoChat2 (Li et al., 2024b)	7B	19.3	1.28	1.98	0.94
ST-LLM (Liu et al., 2024c)	7B	23.8	2.23	2.86	1.13
ShareGPT4Video (Chen et al., 2024d)	16B	26.7	1.85	2.65	1.01
Chat-UniVi-v1.5 (Jin et al., 2023)	13B	25.9	2.07	2.53	1.21
VILA (Lin et al., 2024a)	8B	28.7	1.56	2.89	1.35
LongVA (Zhang et al., 2024)	7B	28.7	2.94	3.86	1.78
LLaVA-OneVision (Li et al., 2024a)	7B	31.1	1.63	1.78	1.08
Video-CCAM (Fei et al., 2024)	14B	29.7	2.63	3.48	1.83
Kangaroo (Liu et al., 2024b)	8B	30.2	2.56	2.81	1.94
VITA (Fu et al., 2024b)	8×7B	33.3	3.06	3.53	2.06
Qwen2-VL (Wang et al., 2024b)	72B	41.3	3.58	5.32	3.31
InternVL2 (OpenGVLab, 2024)	78B	42.2	3.91	5.05	2.64
VideoMind (Ours)	2B	31.0	5.94	8.50	4.02
VideoMind (Ours)	7B	38.4	**7.10**	**9.93**	**4.67**

I. Grounded VideoQA Performance

Table 3: Performance comparison on Grounded VideoQA on ReXTime (Chen et al., 2024c). FT indicates fine-tuning on the target dataset.

Method	Size	FT	R@0.3	R@0.5	mIoU	Acc	Acc@IoU
VTimeLLM (Huang et al., 2024a)	7B	✗	28.84	17.41	20.14	36.16	–
TimeChat (Ren et al., 2024)	7B	✗	14.42	7.61	11.65	40.04	–
LITA (Huang et al., 2024b)	13B	✗	29.49	16.29	21.49	34.44	–
VTimeLLM (Huang et al., 2024a)	7B	✓	43.69	26.13	29.92	57.58	17.13
TimeChat (Ren et al., 2024)	7B	✓	40.13	21.42	26.29	49.46	10.92
VideoMind (Ours)	2B	✗	34.31	22.69	24.83	69.06	17.26
VideoMind (Ours)	7B	✗	**38.22**	**25.52**	**27.61**	**74.59**	**20.20**

II. Video Temporal Grounding Performance

Table 4: Performance comparison on Grounded VideoQA on NExT-GQA (Xiao et al., 2024).

Method	Size	IoU			IoP			Acc@GQA
		R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoP	
FrozenBiLM NG+ (Yang et al., 2022)	890M	13.5	6.1	9.6	28.5	23.7	24.2	17.5
SeViLA (Yu et al., 2023)	4B	29.2	13.8	21.7	34.7	22.9	29.5	16.6
LangRepo (Kahatapitiya et al., 2024)	8×7B	–	12.2	18.5	–	28.7	31.3	17.1
VideoStreaming (Qian et al., 2024b)	8.3B	–	13.3	19.3	–	31.0	32.2	17.8
LLoVi (Zhang et al., 2023a)	1.8T	–	15.3	20.0	–	**36.9**	37.3	24.3
HawkEye (Wang et al., 2024f)	7B	37.0	19.5	25.7	–	–	–	–
VideoChat-TPO (Yan et al., 2024)	7B	41.2	23.4	27.7	47.5	32.8	35.6	25.5
VideoMind (Ours)	2B	45.2	23.2	28.6	51.3	32.6	36.4	25.2
VideoMind (Ours)	7B	**50.2**	**25.8**	**31.4**	**56.0**	35.3	**39.0**	**28.2**

- VideoMind-2B도 강력한 baseline과 경쟁 가능
- VideoMind-7B는 grounding 및 QA 성능에서 높은 성능 달성

II. Grounded VideoQA Performance

Table 5: Performance comparison on video temporal grounding on Charades-STA (Gao et al., 2017) and ActivityNet-Captions (Krishna et al., 2017). FT means fine-tuning on the target dataset.

Method	Size	FT	Charades-STA				ActivityNet-Captions			
			R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
VTimeLLM (Huang et al., 2024a)	7B	✗	51.0	27.5	11.4	31.2	44.0	27.8	14.3	30.4
TimeChat (Ren et al., 2024)	7B	✗	51.5	32.2	13.4	–	–	–	–	–
Momentor (Qian et al., 2024a)	7B	✗	42.6	26.6	11.6	28.5	42.9	23.0	12.4	29.3
ChatVTG (Qu et al., 2024)	7B	✗	52.7	33.0	15.9	34.9	40.7	22.5	9.4	27.2
VideoChat-TPO (Yan et al., 2024)	7B	✗	58.3	40.2	18.4	38.1	–	–	–	–
E.T. Chat (Liu et al., 2024e)	4B	✗	65.7	45.9	20.0	42.3	24.1	12.8	6.1	18.9
Grounded-VideoLLM (Wang et al., 2024a)	4B	✗	54.2	36.4	19.7	36.8	–	–	–	–
TRACE (Guo et al., 2024)	7B	✗	–	40.3	19.4	–	–	–	–	–
LLaVA-ST (Li et al., 2025a)	7B	✗	63.1	44.8	23.4	42.4	–	–	–	–
UniTime (Li et al., 2025b)	7B	✗	–	**59.1**	**31.9**	**52.2**	–	22.8	14.1	27.3
VideoMind (Ours)	2B	✗	67.6	51.1	26.0	45.2	44.0	26.5	12.6	30.1
VideoMind (Ours)	7B	✗	**73.5**	**59.1**	31.2	50.2	**48.4**	**30.3**	**15.7**	**33.3**

- Timestamp decoder + Verifier 조합이 temporal grounding 성능 향상에 기여
- VideoMind-7B는 Charades-STA에서 강한 성능 달성

III. General VideoQA Performance

Table 6: Performance comparison on General VideoQA on Video-MME (Fu et al., 2024a), MLVU (Zhou et al., 2024), and LVBench (Wang et al., 2024c).

Method	Size	Video-MME		MLVU	LVBench
		All	Long	M-Avg	Overall
GPT-4o (OpenAI, 2024a)	–	71.9	65.3	54.5	30.8
Gemini-1.5-Pro (Reid et al., 2024)	–	75.0	67.4	–	33.1
Video-LLaVA (Lin et al., 2023a)	7B	41.1	37.8	29.3	–
TimeChat (Ren et al., 2024)	7B	34.3	32.1	30.9	22.3
MovieChat (Song et al., 2023)	7B	38.2	33.4	25.8	22.5
PLLaVA (Xu et al., 2024)	34B	40.0	34.7	53.6	26.1
VideoChat-TPO (Yan et al., 2024)	7B	48.8	41.0	54.7	–
LongVA (Zhang et al., 2024)	7B	52.6	46.2	56.3	–
VideoMind (Ours)	2B	55.4	46.3	58.7	35.4
VideoMind (Ours)	7B	**58.2**	**49.2**	**64.4**	**40.8**

- VideoMind는 cue segment를 먼저 찾고 답변함으로써 긴 비디오 QA 성능을 개선
- VideoMind-7B가 전반적으로 2B보다 높은 성능

IV.Experiments

Table 7: Performance and efficiency comparison of different test-time scaling and role integration strategies. Mem indicates the peak GPU memory consumption. Notably, Chain-of-LoRA achieves the best performance with minimal memory cost.

Method	Mem	NExT-GQA		Charades-STA		Video-MME	
		mIoU	Acc	R@0.5	mIoU	All	Long
Qwen2-VL-2B	4.1G	–	69.6	–	–	53.0	43.1
+CoT	4.1G	–	69.7	–	–	52.8	43.3
+All-in-One	4.2G	28.0	70.5	47.8	42.1	53.6	43.6
+All-Distributed	16.6G	28.6	71.4	51.1	45.2	55.4	46.3
+Chain-of-LoRA	**4.2G**	**28.6**	**71.4**	**51.1**	**45.2**	**55.4**	**46.3**

- Text-based CoT는 성능 향상이 거의 없음
- All-Distributed는 성능은 좋지만 메모리 사용량이 큼
- Chain-of-LoRA는 All-Distributed와 같은 성능을 내면서 메모리는 크게 절약

IV.Experiments

1. Grounder : 관련 visual cue를 찾으면서 QA 성능 개선

2. Verifier

- candidate moment 선택을 개선
- Charades-STA에서 mIoU 향상

3. Planner

- 모든 sample에 grounding을 강제하지 않고 필요한 경우만 수행
- Grounding 비율을 조절하면서 성능과 효율성 균형 확보

Table 8: Effects of individual roles. A, G, V, P, G% denote the answerer, grounder, verifier, planner, and the percentage of samples processed with the grounder, respectively.

Roles To Use					ReXTime		Charades-STA		
A	G	V	P	G%	mIoU	Acc	R@0.5	R@0.7	mIoU
✓				0%	–	68.0	–	–	–
✓	✓			100%	24.5	68.8	–	–	–
✓	✓	✓		100%	24.8	69.1	–	–	–
✓	✓	✓	✓	100%	24.7	69.2	–	–	–
✓	✓	✓	✓	40%	**26.7**	**70.0**	–	–	–
	✓				–	–	47.2	21.7	42.0
	✓	✓			–	–	**51.1**	**26.0**	**45.2**

V. Conclusion

seed	Audio	Text	detr	LLM	rerank	test_R1@0.5	test_R1@0.7	test_mAP	test_mAP@0.5	test_mAP@0.75
2023	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	x	x	42.09	26.06	20.99	36.59	20.43
2023	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : 2	42.61	26.28	21.08	36.84	20.52
2023	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : ratio 50%	42.69	26.43	21.08	36.83	20.56
2023	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2.5_omni	logit & margin : 2	42.76	26.43	21.14	37	20.55
2024	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	x	x	42.09	26.5	21.18	37.81	19.99
2024	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : 2	42.69	26.65	21.32	38.1	20.09
2024	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : ratio 50%	42.69	26.73	21.32	38.09	20.08
2024	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2.5_omni	logit & margin : 2	42.76	26.87	21.36	38.24	20.12
2025	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	x	x	44.54	26.87	21.12	36.75	20.67
2025	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : 2	44.91	27.02	21.19	36.9	20.68
2025	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : ratio 50%	45.06	27.02	21.23	36.97	20.71
2025	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2.5_omni	logit & margin : 2	45.36	27.1	21.32	37.24	20.75
2026	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	x	x	43.58	26.5	21.69	38.05	20.41
2026	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : 2	43.95	26.73	21.73	38.19	20.44
2026	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : ratio 50%	43.65	26.65	21.76	38.13	20.47
2026	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2.5_omni	logit & margin : 2	43.8	26.58	21.81	38.35	20.48
2027	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	x	x	42.54	25.91	21.26	37.48	20.12
2027	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : 2	42.84	26.06	21.35	37.72	20.14
2027	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2_audio	logit & margin : ratio 50%	42.76	26.06	21.3	37.6	20.13
2027	LAION_CLAP	LAION_CLAP_text	qd_detr_mamba_v6	qwen2.5_omni	logit & margin : 2	43.21	26.28	21.45	37.92	20.22

- AMR 적용 아이디어

1. QD-DETR이 생성한 top-k audio moments를 Verifier-style로 재검증

2. LLM/audio-LLM을 re-ranking verifier로 활용

3. confidence score를 submission ranking에 반영

중앙대학교
CHUNG-ANG UNIVERSITY

감사합니다.

2026-05-21